Entropy-Based Optimization of Wavelet Spatial Filters

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***Abstract*—A new class of spatial filters for surface electromyographic (EMG) signal detection is proposed. These filters are based on the 2-D spatial wavelet decomposition of the surface EMG recorded with a grid of electrodes and inverse transformation after zeroing a subset of the transformation coefficients. The filter transfer function depends on the selected mother wavelet in the two spatial directions. Wavelet parameterization is proposed with the aim of signal-based optimization of the transfer function of the spatial filter. The optimization criterion was the minimization of the entropy of the time samples of the output signal. The optimized spatial filter is linear and space invariant. In simulated and experimental recordings, the optimized wavelet filter showed increased selectivity with respect to previously proposed filters. For example, in simulation, the ratio between the peak-to-peak amplitude of action potentials generated by motor units 20° apart in the transversal direction was 8.58% (with monopolar recording), 2.47% (double differential), 2.59% (normal double differential), and 0.47% (optimized wavelet filter). In experimental recordings, the duration of the detected action potentials decreased from (mean ± SD) 6.9 ± 0.3 ms (monopolar recording), to 4.5 ± 0.2 ms (normal double differential), 3.7 ± 0.2 ms (double differential), and 3.0 ± 0.1 ms (optimized wavelet filter). In conclusion, the new class of spatial filters with the proposed signal-based optimization of the transfer function allows better discrimination of individual motor unit activities in surface EMG recordings than it was previously possible.**

***Index Terms*—Motor unit, selectivity, spatial filtering, surface electromyography (EMG), wavelet design.**

I. Introduction

SURFACE ELECTROMYOGRAPHIC (EMG) signals are usually recorded to obtain global information on the muscle electrophysiological properties. Surface detection systems have poorer spatial selectivity than invasive recordings, thus it is often not possible to separate the contributions of individual motor units from the so-called interference EMG signal. The interference EMG signal is described by global variables, such as amplitude or characteristic spectral frequencies [17], that can be used for an indirect assessment of the underlying physiological mechanisms. Although global EMG analysis has found important applications in applied fields [18], the interpretation of variables extracted from the interference EMG is complicated by many factors of influence [7]. Direct analysis of motor unit properties is preferable but technically difficult.

Methods have been proposed to identify motor unit activities from the surface EMG. These techniques are based on two main categories of recording modalities: 1) spatial filtering [22] and 2) spatial sampling [24]. Spatial filtering consists in the linear combination of signals recorded from 1-D or 2-D grid of detection points [finite impulse response (FIR) filters]. Spatial sampling is the detection of signals from multiple locations over the skin surface to assess the spatial distribution of muscle electrical activity. Advances in hardware design currently allow the amplification and concomitant recording of more than 100 surface EMG signals which provide high-density spatial sampling in combination with high-selective spatial filtering [26]. Spatial filtering reduces the number of motor unit action potentials detected in each location and spatial sampling increases the amount of information for discrimination between shapes of action potentials of different motor units.

Many spatial filters have been proposed in the literature. Most of them, such as the normal double differential (Laplacian) [22] or the second-order inverse binomial [2], are obtained from the field of image processing and represent edge detection filters, i.e., high-pass filters. However, the selectivity of spatial filters depends on the volume conductor properties, thus performance depends on the subject and muscle [8]. In contrast to general filters, not adapted to the recording condition, Farina and Rainoldi [4] proposed a spatial filter designed on the basis of a model of volume conductor. The filter showed better performance than previously proposed filters, although it was tested only on simulated signals and required *a priori* knowledge of volume conductor properties (conductivity and thickness of the tissues). More recently, Östlund *et al.* [21] proposed the adaptation of the spatial filter to the signal properties rather than to the volume conductor properties. The filter was designed to reduce the influence of noisy contacts in the resultant signal and was tested on signals with a single active motor unit and spatially correlated noise. Farina and Merletti [5] and Mesin *et al.* [19] also proposed spatial filters adapted to the signal features with optimization performed to minimize the bias in the estimation of motor unit conduction velocity.

In this paper, we propose a new class of spatial filters, based on the spatial 2-D wavelet decomposition of the surface EMG and inverse transformation after zeroing a subset of the transformation coefficients. The filter transfer function depends on the selected mother wavelet in the two spatial directions. We thus propose wavelet parameterization and optimization of the

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mother wavelet based on the entropy of the output signal. The optimized filter maintains the properties of linearity and space invariance, necessary for practical applications.

II. Methods

A. Wavelet Spatial Filters

In one dimension, the dyadic discrete wavelet transform (DWT) decomposes a signal $f(x)$ on a basis where all the functions are dilated and translated versions of a prototype function ψ, called mother wavelet. The projections of the signal into these basis functions return detail coefficients $d_f(j,k) = \langle f(x), \psi_{j,k}(x)\rangle$, where $\psi_{j,k}(x) = 2^{-j/2}\psi(2^{-j}x - k)$. Approximation coefficients $a_x(j,k)$ are obtained by projecting the signal on dilated and translated versions of the scaling function φ, $a_f(j,k) = \langle f(x), \varphi_{j,k}(x)\rangle$, where $\varphi_{j,k}(x) = 2^{-j/2}\varphi(2^{-j}x - k)$. The set of approximation and detail coefficients can be alternatively computed by Mallat's algorithm with the application of a filter bank [multiresolution analysis (MRA)] [15].

In the MRA framework, the scaling function φ and its associated mother wavelet ψ are related to the filters h and g (used for computation of the DWT through a filter bank) by the two-scale recursive relations $\varphi(x/2) = \sqrt{2}\sum_n h[n]\varphi(x-n)$ and $\psi(x/2) = \sqrt{2}\sum_n g[n]\phi(x-n)$. For orthogonal wavelets, g can be deduced from h from the relation $g[k] = (-1)^{1-k}h[1-k]$, thus h defines ψ.

In two dimensions and for the case of separable wavelets, the MRA implies the definition of a 2-D scaling function $\varphi(x,y)$ and three 2-D wavelet functions $\psi^H(x,y)$, $\psi^V(x,y)$, and $\psi^D(x,y)$

$$\begin{aligned}\varphi(x,y) &= \varphi_1(x)\varphi_2(y)\\ \psi^H(x,y) &= \psi_1(x)\varphi_2(y)\\ \psi^V(x,y) &= \varphi_1(x)\psi_2(y)\\ \psi^D(x,y) &= \psi_1(x)\psi_2(y).\end{aligned} \tag{1}$$

As for the 1-D case, the signal is projected on translated and scaled versions of the scaling and wavelet functions to obtain approximation $a_{j,m,n}$ and detail $d^i_{j,m,n}$ coefficients

$$\begin{aligned}a_{j,m,n} &= \langle f(x,y), \varphi_{j,m,n}(x,y)\rangle\\ d^i_{j,m,n} &= \left\langle f(x,y), \psi^i_{j,m,n}(x,y)\right\rangle\end{aligned} \tag{2}$$

with

$$\begin{aligned}\varphi_{j,m,n}(x,y) &= 2^{j/2}\varphi(2^jx-m, 2^jy-n)\\ \psi^i_{j,m,n}(x,y) &= 2^{j/2}\psi(2^jx-m, 2^jy-n), \qquad i=\{V,H,D\}.\end{aligned} \tag{3}$$

Two filter pairs h_1 and h_2 and g_1 and g_2 define the 2-D wavelet decomposition which, in case of separable wavelets, can be seen as the application of a 1-D transform in each direction. Wavelet decomposition provides a subband analysis of the signal. In two dimensions, the first level of decomposition provides four subbands, corresponding to the projection of the signal on the scaling function (low frequency in both directions) and on the three wavelet functions (low and high frequencies or high frequencies in both directions).

The operation of zeroing the wavelet coefficients in specific subbands and inverse transforming corresponds to filtering. We propose a class of spatial filters for surface EMG detection based on zeroing a subset of coefficients of the 2-D spatial wavelet decomposition of the monopolar EMG signals recorded with an electrode grid. For each instant of time, the monopolar EMG signals are decomposed with a 2-D wavelet transform. The inverse transformation is applied after zeroing some of the transformation coefficients. The central spatial location after inverse transformation is then used as the filter output. These filters are linear and space invariant and depend on the selection of the mother wavelets and of the coefficients to be zeroed.

B. Parameterization of the Mother Wavelet

In one dimension and in the case of orthogonal wavelets, h defines the scaling and wavelet functions. However, to generate an orthogonal MRA wavelet, h must satisfy some constraints. For an FIR filter of length L, there are $L/2+1$ sufficient conditions to ensure the existence and orthogonality of the scaling function and wavelet [12], [13]. Thus, $L/2-1$ degrees of freedom remain to design the filter h. The lattice parameterization described by Vaidyanathan [25] offers the opportunity to design h via unconstrained optimization: the L coefficients of h can be expressed in term of $L/2-1$ new free parameters. For example, if $L=4$, one single parameter α is needed and h is given by [1], [23]

$$\begin{aligned}h[0] &= (1-\cos(\alpha)+\sin(\alpha))/(2\sqrt{2})\\ h[1] &= (1+\cos(\alpha)+\sin(\alpha))/(2\sqrt{2})\\ h[2] &= (1+\cos(\alpha)-\sin(\alpha))/(2\sqrt{2})\\ h[3] &= (1-\cos(\alpha)-\sin(\alpha))/(2\sqrt{2}).\end{aligned} \tag{4}$$

In case of 2-D wavelet decomposition with separable and orthogonal wavelets, the filters h_1 and h_2 should satisfy the same conditions as in the case of the 1-D transformation and can be parameterized in the same way. In this case, for filter lengths L, $L-2$ free parameters, double with respect to the 1-D case, are necessary to define the filter pair h_1 and h_2.

C. Optimization Criterion

The class of wavelet spatial filters defined previously depends on $L-2$ free parameters and on the coefficients to be zeroed. This provides an infinite number of filters (class of spatial filters) with different selectivity properties. It is hypothesized that the best filter depends on the signal, and therefore, a signal-based selection of the optimal spatial filter and an optimization criterion are necessary. For the purpose of reducing the relative contribution of small motor units in surface EMG recordings, the criterion should reflect the capacity of the filter to enhance a few motor unit activities and attenuate the contribution of the others.

For surface EMG recordings, selectivity in the transverse and depth directions can be quantified from the ratio between amplitudes of action potentials generated by a superficial motor unit under the recording electrodes and a motor unit displaced in the transverse direction or deeper [8]. The larger the attenuation in amplitude with increasing transversal displacement or depth is, the larger the selectivity becomes. In the longitudinal direction, selectivity can be quantified from the duration of the recorded

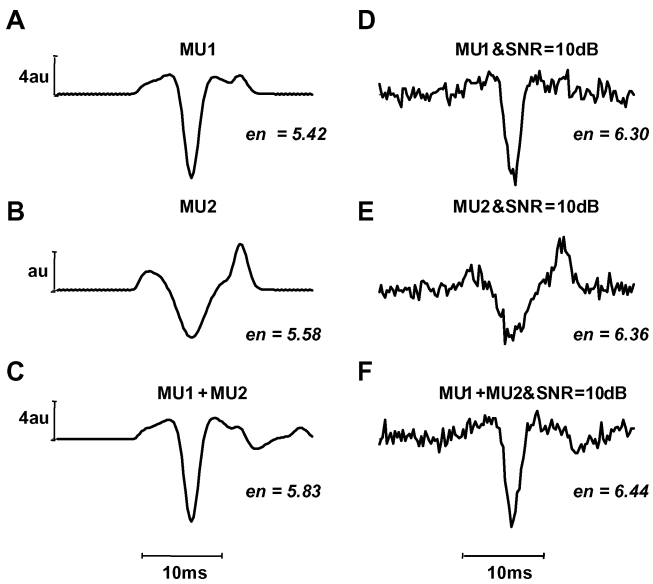

Fig. 1. Entropy in representative simulated signals. Entropy increases when computed from a single motor unit action potential of shorter (a) with respect to longer duration (b). It further increases when the two action potentials are concomitantly present in the recording (c) and when noise is added (d–f). Thus, minimal entropy corresponds to the minimal number of action potentials, with shortest duration and minimal noise (au: arbitrary units; en: entropy).

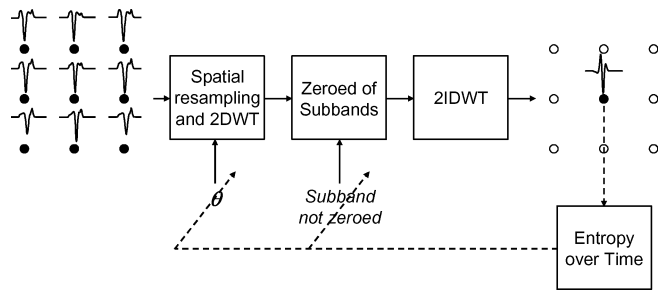

Fig. 2. Schematic representation of the optimization procedure for the wavelet spatial filters. For each instant of time, the signal in the two spatial coordinates is decomposed through a 2-D wavelet transform (separable and orthogonal wavelets in this paper). The decomposition at the first level provides four subbands. The filtering operation consists in zeroing three of the four subbands and inverse transforming. The best combination of subband and mother wavelet is selected through the entropy of the output signal (from the central electrode) in the time domain.

action potential, with larger selectivity for smaller durations [8]. In addition to high selectivity, a spatial filter should maximize the output signal-to-noise ratio (SNR).

According to the previous quantification of selectivity and considering a surface EMG recording made of contributions from a number of active motor units, the filter output should have only a small number of samples different from zero, corresponding to a few short motor unit action potentials. This condition can be quantified by the entropy of the samples of the output signal (Fig. 1)

$$En = -\sum_{i=1}^{N} p(i) \log_2 p(i) \tag{5}$$

where N is the number of samples and $p(i)$ is the square of the absolute value of the ith signal sample normalized by the summation of the squares of the absolute values of all samples. If more than one sample is different from zero, minimization of entropy corresponds to the minimum number of nonzero samples and to the largest difference between the nonzero samples. If the recording is made of a series of different action potentials, filter outputs with low entropy correspond to the maximization of difference in amplitude among action potentials and the minimization of duration of the action potentials (minimum number of nonzero samples). These concepts are shown in Fig. 1. The minimization of the entropy of the output signal is thus proposed as optimization criterion for the selection of the optimal (according to this criterion) spatial filter within the defined class of wavelet spatial filters.

D. Optimization Method

In this study, the two wavelet functions in the two spatial directions have been chosen equal $(h_1 = h_2 = h)$ and the filter length has been fixed to $L = 4$. Thus, the total number of free parameters for wavelet design is 1. Extension of the method to more free parameters can be obtained from the lattice parameterization [25] (formulas reported in [1] and [23]). The spatial decomposition is performed for only one level, leading to four subbands. Three subbands are zeroed and one is left for inverse transformation. Extension of the method to include more decomposition levels and/or a larger selection of subbands implies a deeper decomposition of the 2-D signal. With the choices on filters and decomposition levels, the optimal spatial filter is obtained by minimizing the entropy of the output signal over 1) the free parameter that defines the mother wavelet and 2) the subband that is not zeroed.

The following steps define the optimization procedure (Fig. 2).

1) Select a value for the parameter α [(4)]
   a) compute the filters $h_1^\alpha = h_2^\alpha = h^\alpha$ using (4) [23].
2) For each time instant
   a) decompose the signal via the 2-D DWT derived from h^α according to Mallat's algorithm stopped at the first decomposition level (four subbands in two dimensions);
   b) for each of the four subbands
      — null the coefficients in the other subbands;
      — compute the inverse 2-D DWT.
3) Compute the entropy of the signal samples of the resultant central signal of the grid (output signal) for each preserved subband.
4) Repeat steps 1)–3) for all values of α in a predefined set of values.
5) Choose the parameter α^{opt} and the subband which lead to the minimum entropy of the output signal.

E. Simulated Signals

To test the proposed spatial filter, simulated signals were generated with the model proposed in [9]. The model consists in the description of the volume conductor as cylindrical with four tissue layers (bone, muscle, subcutaneous, and skin) of different conductivity. Two conditions were simulated, corresponding to signal recordings of 200 ms containing one or two motor unit action potentials. In both conditions, the motor units had 200 fibers, circular territory, and fiber density 20 fiber/mm^2. The detection system consisted of a 3 × 3 electrode grid with 5-mm

interelectrode distance in both directions. In the first set of simulations, the simulated motor unit was located 1) directly under the center of the recording system at a depth of 1 mm into the muscle, 2) 5 mm into the muscle, or 3) at 20° of radial distance and 1-mm depth into the muscle. Subcutaneous layer thickness was set to 1.5 mm. To test the sensitivity of the filters to noise, noise was added with 10-dB SNR.

The second set of simulations involved two motor units. A reference motor unit was located directly under the recording system at a depth of 1 mm into the muscle and a second motor unit was located 1) at 5-mm depth into the muscle or 2) at 20° transversal distance from the recording system and 1-mm depth into the muscle. The time instants, where the motor unit action potentials were located in the 200-ms recordings, were randomly selected in 20 simulations for each condition, allowing full superposition of the two action potentials.

F. Experimental Signals

Experimental signals were collected from the tibialis anterior muscle of four healthy subjects (three males, one female, age mean ± SD 27.5 ± 2.1 years) with a 2-D grid of 5 × 13 electrodes (8-mm interelectrode distance in both directions). Only the lowest, central 3 × 3 part of the grid was used for subsequent analysis. This part of the grid was located between the most distal innervation zone, identified as previously described [16], and the distal tendon. After determination of the maximal voluntary contraction (MVC) force, the subjects were asked to perform an ankle dorsi-flexion at 5% MVC for 1 min. Monopolar surface EMG were amplified 5000 times (64-channel surface EMG amplifier, SEA 64, LISiN-OT Bioelettronica, Torino, Italy; −3 dB bandwidth 10–500 Hz), sampled at 2048 Hz, and converted to digital form by a 12-bit analog-to-digital (A/D) converter.

G. Signal Analysis

The simulated and experimental signals consisted of only 3 × 3 electrodes, which is currently the most common recording size for 2-D EMG spatial filters [22]. This choice was adopted for comparison of the proposed method with previously proposed spatial filters. Spatial wavelet decomposition should however be applied to a larger number of spatial samples due to the length of the scaling and wavelet filters ($L = 4$ in this study). In practice, this can be achieved by a smaller interelectrode distance and a larger number of electrodes. However, for comparison purposes, in this study, the wavelet filter was applied to the 3 × 3 recordings with interpolation by a factor 2. In all cases, the values of the free parameter α for wavelet optimization varied from 0 to 2π with increments $\pi/100$.

The following recording systems were compared with the optimized wavelet filter: monopolar, single differential, double differential, normal double differential [22], and inverse binomial of order 2 [2]. The previously proposed recording systems were applied to both the original 3 × 3 recordings and to the 3 × 3 signals in the center of the spatially interpolated recordings but the results were comparable, thus for the classic filters results are reported on the original signals. In order to assess the need for wavelet optimization, the results of the optimized wavelet filter were also compared with those obtained by the same filter with Daubechies wavelet, with the same analysis filter length as for the optimized wavelet.

TABLE I
DURATION (ESTIMATED AS THE RATIO BETWEEN AREA AND PEAK-TO-PEAK VALUE AND EXPRESSED IN MILLISECONDS) OF AN ACTION POTENTIAL DETECTED BY THE TESTED SPATIAL FILTERS AND GENERATED BY A MOTOR UNIT WITH 200 FIBERS. THE MOTOR UNIT IS LOCATED IN THREE POSITIONS INTO THE MUSCLE (SEE TEXT FOR DETAILS). MON: MONOPOLAR RECORDING, SD: SINGLE DIFFERENTIAL, DD: DOUBLE DIFFERENTIAL, NDD: LAPLACIAN, IB2: INVERSE BINOMIAL OF ORDER 2, DB2: WAVELET FILTER WITH DAUBECHIES MOTHER WAVELET, OWSF: OPTIMIZED WAVELET FILTER

Duration (ms)	MU1	MU2	MU3
Filters	Depth (1mm)	Transversal Distance (20degrees)	Depth (5mm)
MON	2.85	2.96	3.31
SD	1.78	3.87	3.01
DD	1.78	2.55	2.99
NDD	1.86	2.40	3.38
IB2	1.90	2.39	3.43
DB2	1.70	2.51	2.57
OWSF	1.55	1.43	1.98

The selectivity of the spatial filters in the first set of simulations with only one active motor unit was evaluated from the duration of the output action potential [8]. Duration was computed as the ratio between area and peak-to-peak value of the output signal in the 200-ms window. The effect of the filter transfer function on additive noise was assessed by computing the SNR at the output of the filter. For this purpose, the filters were applied separately to the signal and noise part of the recording since all filters are linear. In case of optimized wavelet filter, the optimal wavelet was selected using the noisy recording and the resultant filter applied separately to the signal and noise, as for the other filters. In case of simulations with two motor units, selectivity was assessed as the ratio (expressed in percentage) between the peak-to-peak values of the action potentials of the two motor units at the output of the spatial filters.

Six motor unit action potentials were visually identified from the experimental EMG recordings of each subject using the Laplacian filter output [6]. Thus, 24 motor unit action potentials were analyzed in total. The ratio between area of the signal and peak-to-peak amplitude in a window of 40 ms, symmetrical with respect to the peak of the detected action potential, provided an estimation of the duration of the action potentials. A 40-ms window duration was chosen to limit the contributions of action potentials other than the selected one in the calculation of duration. Shorter duration was associated to higher selectivity [6], as for the simulated potentials. A one-way repeated measure analysis of variance (ANOVA) with Student–Newman–Keuls (SNK) *post hoc* test was used to assess dependency of this ratio on the spatial filter.

III. RESULTS

A. Simulated Signals

Table I reports the duration of the output action potential (one motor unit simulated) for the filters investigated and the three locations of the motor unit. The proposed filter resulted in the shortest duration of the output action potential among the filters

TABLE II
SNR (MEAN± SD) AT THE OUTPUT OF THE SPATIAL FILTER IN CASE OF SIMULATION OF A SINGLE MOTOR UNIT ACTION POTENTIAL WITH INPUT (FOR THE MONOPOLAR RECORDING) SNR OF 10 dB (200 NOISE REALIZATIONS). THE NOISE WAS TEMPORALLY AND SPATIALLY WHITE. MON: MONOPOLAR RECORDING, SD: SINGLE DIFFERENTIAL, DD: DOUBLE DIFFERENTIAL, NDD: LAPLACIAN, IB2: INVERSE BINOMIAL OF ORDER 2, DB2: WAVELET FILTER WITH DAUBECHIES MOTHER WAVELET, OWSF: OPTIMIZED WAVELET FILTER

SNR	MU1	MU2	MU3
Filters	Depth (1mm)	Transversal Distance (20degrees)	Depth (5mm)
MON	10.01 ± 0.21	9.95 ± 0.15	9.98 ± 0.20
SD	8.76 ± 0.16	4.54 ± 0.17	5.85 ± 0.14
DD	7.83 ± 0.24	2.96 ± 0.13	2.78 ± 0.16
NDD	7.24 ± 0.23	2.75 ± 0.15	2.64 ± 0.20
IB2	7.50 ± 0.22	3.08 ± 0.15	3.16 ± 0.19
DB2	11.93 ± 0.17	13.84 ± 0.15	13.26 ± 0.19
OWSF	10.42 ± 0.38	13.88 ± 1.53	13.62 ± 0.19

TABLE III
RATIO (PERCENT) BETWEEN THE PEAK-TO-PEAK AMPLITUDE OF ACTION POTENTIALS GENERATED BY TWO MOTOR UNITS FOR THE TESTED SPATIAL FILTERS (SEE TEXT FOR DETAILS ON THE LOCATION OF THE MOTOR UNITS). MON: MONOPOLAR RECORDING, SD: SINGLE DIFFERENTIAL, DD: DOUBLE DIFFERENTIAL, NDD: LAPLACIAN, IB2: INVERSE BINOMIAL OF ORDER 2, DB2: WAVELET FILTER WITH DAUBECHIES MOTHER WAVELET, OWSF: OPTIMIZED WAVELET FILTER. THE RESULTS FROM DB2 AND OWSF ARE REPORTED AS MEAN± SD SINCE THE OPTIMIZATION OF THE SUBBAND (FOR DB2) AND OF THE SUBBAND AND MOTHER WAVELET (OWSF) DEPENDS ON THE POSITIONS OF THE ACTION POTENTIALS IN THE 200-ms RECORDINGS (20 RANDOM LOCATIONS WERE SIMULATED)

Ratio (%)	MU2	MU3
Filters	Transversal Distance(20degrees)	Depth(5mm)
MON	8.58	22.81
SD	1.95	8.63
DD	2.47	5.39
NDD	2.59	5.79
IB2	2.67	6.14
DB2	1.91 ± 0.24	3.04 ± 0.01
OWSF	0.47 ± 0.01	3.22 ± 0.01

tested in all conditions, with substantial improvement also with respect to the same filter with Daubechies wavelet.

Table II reports the output SNR in case of one simulated action potential. The monopolar recording was the reference with 10-dB SNR. All spatial filters, except the wavelet filters, determined a reduction in the SNR. The wavelet spatial filter improved the output SNR with respect to the input. Similar results were obtained with Daubechies and optimized wavelet.

Table III shows the results from the simulations with two motor units and Fig. 3 reports the effect of wavelet selection on the ratio performance index. The reduction of the second source with the wavelet filters was substantially larger than with the other filters in both conditions analyzed (Fig. 4) and depended on the selected wavelet (Table III and Fig. 3).

When assessing selectivity, the optimal subband, according to the criterion on entropy, was in almost all simulated cases corresponding to high frequencies in both directions. Thus, the resulting spatial filter was high pass in the two spatial directions, as the Laplacian spatial filter. This result was expected since high-pass filters compensate for the low-pass transfer function of the volume conductor [22]. For noisy signals, however, the optimal subband selected with the entropy criterion varied among conditions since it was a tradeoff between selectivity and noise reduction.

B. Experimental Signals

Fig. 5 shows an example of experimental single motor unit action potential filtered by the analyzed spatial filters. The performance parameter based on the ratio between area and peak-to-peak value was significantly affected by the spatial filter (ANOVA: $F = 86.2$, $P < 0.001$). The SNK *post hoc* test revealed that the ratio was the smallest for the optimized wavelet filter with respect to all the others ($P < 0.05$; monopolar: 6.9 $\pm$ 0.3 ms; single differential: 4.1 $\pm$ 0.2 ms; double differential: 3.7 $\pm$ 0.2 ms; normal double differential: 4.5 $\pm$ 0.2 ms; second-order inverse binomial: 4.3 $\pm$ 0.2 ms; wavelet filter with Daubechies wavelet: 3.4 $\pm$ 0.2 ms; optimized wavelet filter: 3.0 $\pm$ 0.1 ms). Fig. 6 shows an example of application of the optimized wavelet spatial filter to a longer signal portion. The optimal wavelet depends on the signal length. An optimized wavelet spatial filter can be determined from a relatively short signal portion in order to reduce the computational time and maintained for the rest of the signal recording.

The optimal subband for experimental signals was in almost all cases corresponding to high frequencies in both directions, in agreement with the simulation results and the theory of volume conductor.

IV. DISCUSSION

A new class of spatial filters based on spatial wavelet decomposition of the EMG signal has been proposed. The filter transfer function was optimized with a criterion that does not require *a priori* anatomical information. The optimized wavelet spatial filter was superior to previously proposed filters in both simulated and experimental conditions.

A. Spatial Filtering of Surface EMG

The selection of the spatial filter in surface EMG detection is a complex task since spatial selectivity depends on the properties of the volume conductor [8]. Thus, a spatial filter may be more or less selective than another depending on the experimental condition. Spatial filters not adapted to the signal are not optimal in all situations. A better approach is the selection of the spatial filter on a signal basis, maximizing performance with an optimization criterion. The approach of signal-based spatial filtering for EMG detection was proposed by Östlund *et al.* [21], who selected the weights of the spatial filter by solving an eigenvalue problem for kurtosis maximization. The approach has been proven to be effective for enhancing a single main motor unit activity in spatially colored noise [21]. However, the maximization of kurtosis does not correspond to reduction in the number of action potentials and thus the filter does not provide advantages in case of low noise levels [21]. In this study, we proposed a new way of designing EMG spatial filters with the purpose of minimizing the number and duration in time of the detected action potentials.

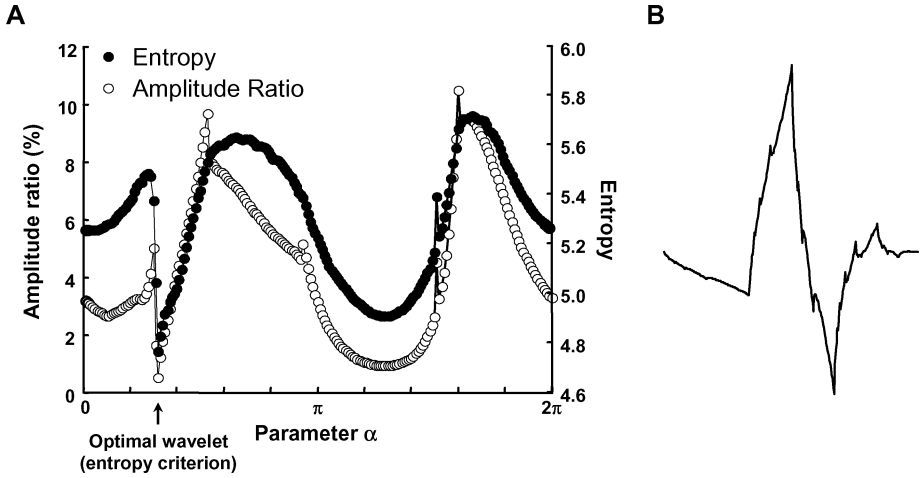

Fig. 3. (a) Entropy of the output signal (criterion for optimization) and ratio (percent) between the peak-to-peak amplitude of action potentials generated by two motor units 20° apart (the same conditions as in Table III), as a function of the parameter α that defines the mother wavelet. Note the variation (from $\sim$0% to $\sim$10%) in the ratio index depending on the mother wavelet, which indicates the importance of optimization, and the close resemblance of the entropy and ratio functions, which indicates that the entropy criterion is associated to the reduction of one action potential with respect to the other (see also Fig. 1). (b) Optimal wavelet is that resulting in minimal entropy (criterion) and in this example is also the wavelet leading to the minimal ratio.

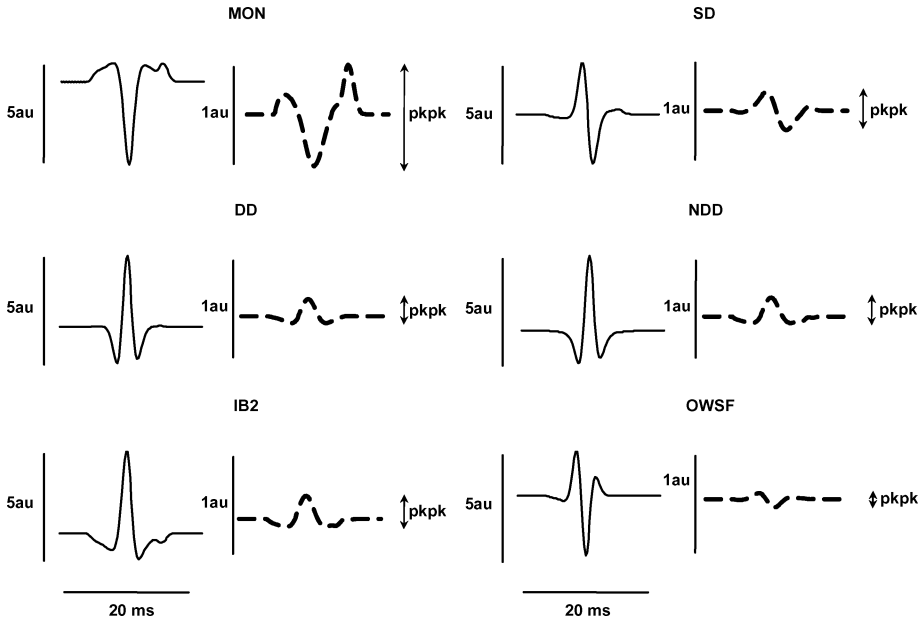

Fig. 4. Simulations with two active motor units. A reference motor unit, whose action potential is shown in solid line, was located under the recording system at a depth of 1 mm into the muscle and a second motor unit, whose action potential is shown in dashed line, was located at 5-mm depth into the muscle at the same transversal location as the first unit. The output signal for the tested spatial filters is shown separately for the two action potentials (solid and dashed lines) but the optimal filter was computed from a 200-ms-long recording with both action potentials. Note the different degree of reduction of the secondary source (dashed line) depending on the filter. MON: monopolar recording, SD: single differential, DD: double differential, NDD: Laplacian, IB2: inverse binomial of order two, DB2: wavelet filter with Daubechies mother wavelet, OWSF: optimized wavelet filter, pkpk: peak-to-peak amplitude; au: arbitrary units.

Spatial filtering reduces the spatial bandwidth of the recorded surface action potential and thus the information contained in the shape of the action potential. However, once the action potentials of individual motor units are identified, their times of occurrence can be used to trigger the averaging of the monopolar derivation, as described in [3]. In this way, spatial filtering is applied to identify individual motor unit discharges and this information is used for both investigating motor unit control strategies and estimating the shape of the monopolar action potential through spike-triggered averaging.

B. Optimization of Wavelet Spatial Filters

Wavelet decomposition allows a subband analysis of the signal and thus the removal of specific frequency components. The transfer function of the resultant filter can be adapted to the signal properties by choosing the mother wavelet and the subbands. This allows the optimization of both the frequency content of the output signal (subband) and the shape of the transfer function (mother wavelet). Mother wavelet optimization requires parameterization. We have previously

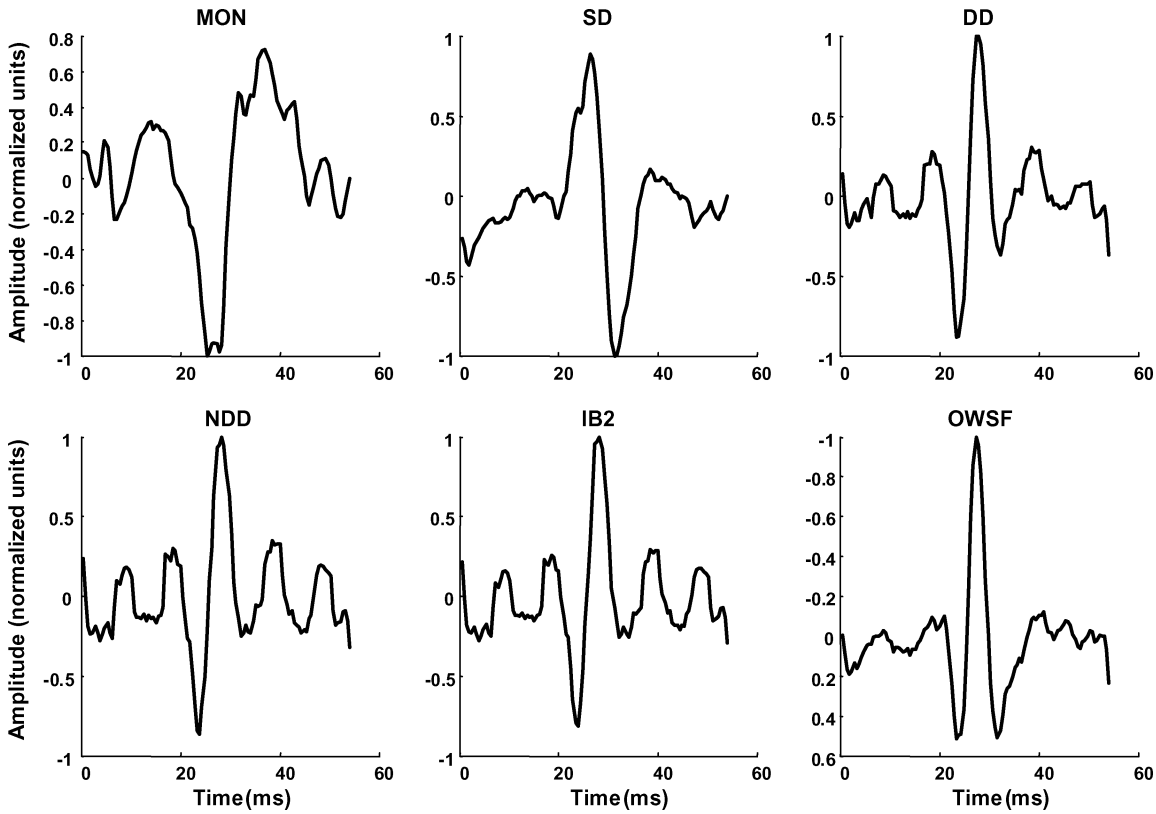

Fig. 5. Examples of experimental surface action potentials detected with the tested filters. MON: monopolar recording, SD: single differential, DD: double differential, NDD: Laplacian, IB2: inverse binomial of order two, DB2: wavelet filter with Daubechies mother wavelet, OWSF: optimized wavelet filter.

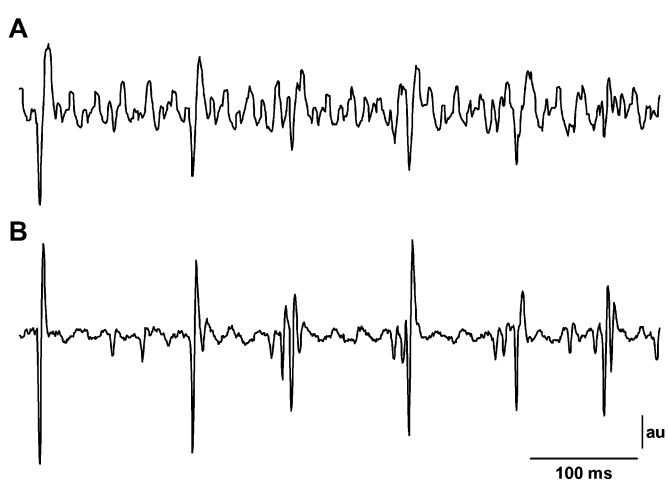

Fig. 6. The 500-ms experimental signal portion (a) detected in monopolar configuration and after the application of the optimized wavelet filter (b). The optimized wavelet filter has been obtained by computing the entropy of the output signal over the entire 500-ms window. au: arbitrary units.

used wavelet parameterization for other biomedical signal processing applications, including classification [11], [14], blind source separation [10], and signal compression [20]. For the purpose of enhancing filter selectivity, the entropy of the output signal time samples was chosen as optimization criterion. When not all samples are zero, minimal entropy corresponds to the minimum number of nonzero samples and the largest difference between nonzero samples. The use of this criterion for wavelet filter selection was successful in reducing the contribution of motor units other than the main one. The optimal wavelet was superior to a classic wavelet (Daubechies) in most simulated conditions and when applied to experimental signals.

The method was applied after spatial interpolation of the signals in order to increase the number of spatial samples. Interpolation can be avoided if the number of electrode is increased. This can be achieved with current recording methods for surface EMG that consist of hundreds of electrodes [26], however this study was limited to the analysis of 3×3 electrode configurations for comparison with previous approaches. The spatial sampling frequency was in this study 125 cycles/m (inverse of the interelectrode distance) which is close or slightly below the Nyquist limit for surface EMG [22]. Moreover, the interpolation (and the DWT) may have presented edge effects due to the limited spatial window of observation. A larger number of more closely spaced electrodes, which is technically feasible, would likely improve performance in future applications.

There are many other potential improvements of the proposed approach. The mother wavelet can be different in the two spatial directions with optimization of two wavelet functions. In this case, with filter length L in both directions, the number of degrees of freedom is $L-2$, double with respect to the case of the same wavelet in both directions. This would increase the number of degrees of freedom in the optimization procedure, and thus increase the computational time, but allow the generation of a

larger number of transfer functions. Since the properties of the volume conductor are different in the two spatial frequency coordinates [9], it is expected that a separate optimization of the wavelet function in the two directions provides a better tradeoff between longitudinal and transverse selectivity. Moreover, if the decomposition is applied to a larger grid, a larger number of decomposition levels can be achieved with better definition of the optimal transfer function. In this study, the optimal subband was in the majority of cases corresponding to high frequencies in both directions, as expected for optimizing selectivity [22], however the optimal transfer function may be asymmetrical when more decomposition levels are chosen. The optimization criterion can also be modified in order to select specific motor units. Finally, the search for the optimal wavelet is currently done by application of all wavelet filters corresponding to a sampling of the free parameters and selection of the best output while this may be implemented more efficiently with gradient methods on the entropy as a function of the free parameters or other minimization procedures.

The potential improvements to the method relate to the many degrees of freedom in the optimization procedure. The main aim of this study was the proposal of the class of wavelet spatial filters for EMG detection. This class can be adapted to the signal features using parameterization of the mother wavelet in different ways, including nonorthogonal wavelets. This study shows the potential of one of these optimization methods.

V. Conclusion

A new class of spatial filters based on wavelet decomposition has been proposed. A specific filter from this class can be selected by optimization of the frequency subband and mother wavelet for minimization of the entropy of the output signal. This increased the selectivity of surface EMG recordings with respect to previously proposed spatial filters in both simulated and experimental signals.

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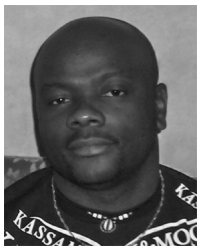

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